

I: excess concentrated/conc. H_2SO_4 , 170°C

III: NaCN/KCN , ethanol, heat under reflux

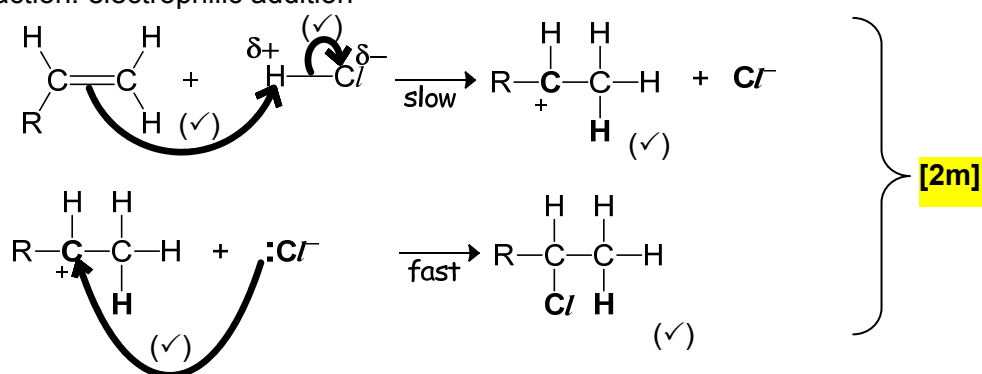
IV: $\text{H}_2\text{SO}_4(\text{aq})/\text{HCl}(\text{aq})$, heat under reflux

[1m]

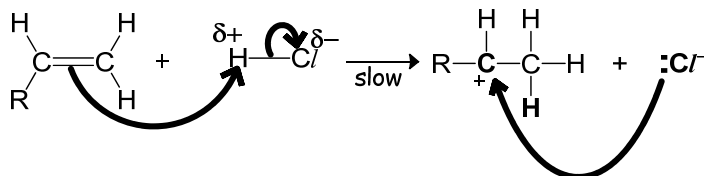
[1m]

[1m]

1 (a) (ii) Type of reaction: electrophilic addition

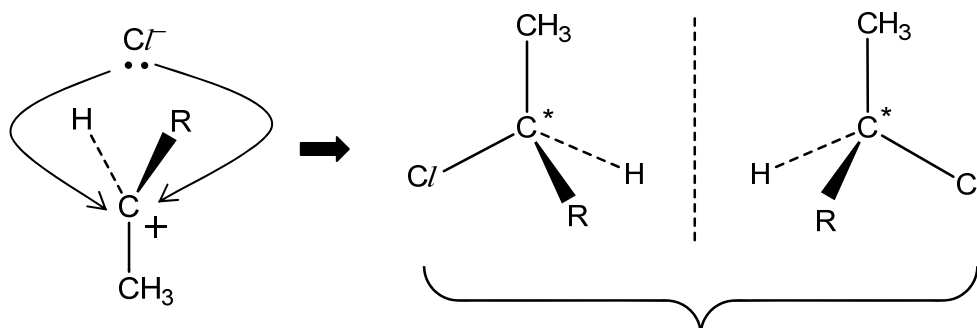


1 (a) (iii)



$\text{RCH}=\text{CH}_2$ undergoes electrophilic addition with HCl to form an organic product, **$\text{RCH}(\text{Cl})\text{CH}_3$** , which has **a chiral centre**.

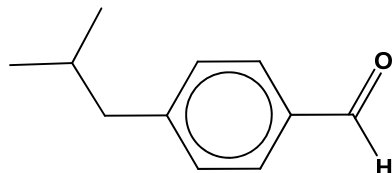
As there is **equal probability for Cl^- to attack either side of the trigonal planar C^+ carbocation**, a **racemic mixture** (OR **equal amounts of optical isomers/enantiomers**) of $\text{RCH}(\text{Cl})\text{CH}_3$ is formed. [1m]



equal amounts of non-superimposable mirror images

1 (a) (iv) The **bulky, non-polar hydrocarbons** in ibuprofen **hinders the formation of effective hydrogen bonds** with water molecules. [1m]

Hence, ibuprofen is not very soluble in water.

1 (b) (i) W:

[1m]

X: CH_3NH_2

[1m]

1 (b) (ii) Z, an aromatic amine, is the weaker base.

[1m]

The p-p orbital overlap results in the delocalisation of the lone pair of electrons on N atom of Z into the π electron system of benzene ring and thus making the lone pair less available for protonation. [1m]

1 (b) (iii) Test: Add $\text{Br}_2(\text{aq})$ to each compound separately. [1m]

Observation: For Z, red-brown $\text{Br}_2(\text{aq})$ decolourises and white precipitate is formed but not for 1-(4-isobutylphenyl)ethanamine. [1m]

1 (c) (i) For order w.r.t. $[\text{C/O}^-]$:Comparing Expt 1 & 2, $[\text{NH}_3]$ remains constant. $\text{Rate} \propto [\text{C/O}^-]$

When $[\text{C/O}^-]$ is halved (or increased to twice or two times its original value) while, the rate of reaction halved (or increased to twice or two times its original value). Thus the order of reaction w.r.t. C/O^- is 1. [1m]

For order w.r.t. $[\text{NH}_3]$:Comparing Expt 1 & 3: $\text{Rate} \propto [\text{C/O}^-]^1 [\text{NH}_3]^y$

$$\frac{4.8 \times 10^{-11}}{4.0 \times 10^{-12}} = \frac{(3.0 \times 10^{-4})^y (4.0 \times 10^{-4})^1}{(1.0 \times 10^{-4})^y (1.0 \times 10^{-4})^1}$$

$$12 = 4(3)^y \Rightarrow y = 1$$

Solving, $y = 1$. Order of reaction w.r.t. NH_3 is 1. [1m](ii) $\text{Rate} = k[\text{NH}_3][\text{C/O}^-]$ [1m]

Using values from expt 1,

$$4.0 \times 10^{-12} = k (1.0 \times 10^{-4}) (1.0 \times 10^{-4})$$

$$k = \underline{4.00 \times 10^{-4} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}} \quad [2m]$$

1 (c) (iii) From the slow step of the mechanism (Step 2),

$$\underline{\text{Rate} = k [\text{NH}_2\text{Cl}] [\text{OH}^-]} \quad (\checkmark)$$

From the equation in step 1 : $[\text{NH}_3] [\text{C/O}^-] \propto [\text{NH}_2\text{Cl}] [\text{OH}^-]$ ($\checkmark$)Thus, $\text{Rate} = k_{\text{overall}} [\text{NH}_3] [\text{C/O}^-]$ ($\checkmark$)3($\checkmark$): [1m]

Alternatively, it can be proven mathematically:



$$K_c = \frac{[\text{NH}_2\text{Cl}][\text{OH}^-]}{[\text{NH}_3][\text{ClO}^-]} \Rightarrow [\text{NH}_2\text{Cl}][\text{OH}^-] = K_c [\text{NH}_3][\text{ClO}^-] (\checkmark)$$

Rate = $k [\text{NH}_2\text{Cl}][\text{OH}^-]$ ($\checkmark$) from the slow step

Rate = $k_{\text{overall}} [\text{NH}_3][\text{ClO}^-]$ ($\checkmark$) where $k_{\text{overall}} = k \cdot K_c$

3($\checkmark$): [1m]

2. (a) (i) DEA **ionises partially** in aqueous solution by **accepting a proton**. [1m]

2. (a) (ii) $\text{pOH} = 14 - 11.2 = 2.8$

$$[\text{OH}^-] = 10^{-2.8} = \underline{1.58 \times 10^{-3}} \text{ mol dm}^{-3}$$

DEA $\equiv \text{H}^+$ (DEA is a monoprotic base)

$$[\text{DEA}] = \frac{25.00 \times 0.050}{20.0} = \underline{0.0625} \text{ mol dm}^{-3}$$

[1m] ans + 3sf

Since $[\text{OH}^-] \ll [\text{DEA}]$, DEA is a **weak base that ionises partially in water**. [1m]

2. (a) (iii) $\text{R}_2\text{NH}(\text{aq}) + \text{H}_2\text{O}(\text{l}) = \text{R}_2\text{NH}_2^+(\text{aq}) + \text{OH}^-(\text{aq})$

$$K_b = \frac{[\text{R}_2\text{NH}_2^+][\text{OH}^-]}{[\text{R}_2\text{NH}]} \quad [1\text{m}]$$

2. (a) (iv) $\underline{\text{R}_2\text{NH}_2^+ + \text{H}_2\text{O} = \text{R}_2\text{NH} + \text{H}_3\text{O}^+}$

Or $\underline{\text{R}_2\text{NH}_2^+ = \text{R}_2\text{NH} + \text{H}^+} \quad [1\text{m}]$

2. (b) (i) $\underline{\text{R}_2\text{NH}(\text{aq})}$, $\underline{\text{R}_2\text{NH}_2^+(\text{aq})}$ and $\text{OH}^-(\text{aq})$ [1m]

2. (b) (ii) $\text{pOH} = \text{p}K_b + \lg \frac{[\text{salt}]}{[\text{base}]}$

$$= \text{p}K_b + \lg \frac{[\text{R}_2\text{NH}_2^+]}{[\text{R}_2\text{NH}]}$$

$$= -\lg(3.98 \times 10^{-5}) + \lg \left(\frac{10.00 \times 0.050}{15.00 \times 0.050} \right) \quad [1\text{m}] \text{ for } [\text{salt}]/[\text{base}] \text{ or volume ratio}$$

$$= 4.22$$

$$\text{pH} = 14 - 4.22 = \underline{9.78} \quad [1\text{m}] \text{ 3 sf}$$

(iii) When a drop of NaOH is added, **OH⁻ is removed by R₂NH₂⁺**.

OR $\underline{\text{R}_2\text{NH}_2^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{R}_2\text{NH}(\text{aq}) + \text{H}_2\text{O}(\text{l})}$ *full arrow must be used

Hence, **pH** of resultant solution **remains approximately unchanged**.

[1m]

- 2 (c) (i) Boiling point of **TEA** is higher than that of DEA. [1m]

1st (✓): Both TEA and DEA have simple molecular structure.

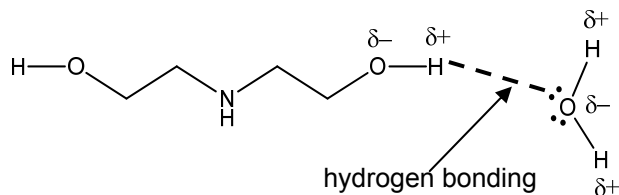
2nd (✓): Due to the greater number of electrons per TEA molecule, more energy is needed to overcome the stronger van der Waals' forces between TEA molecules than those between DEA molecules.

OR

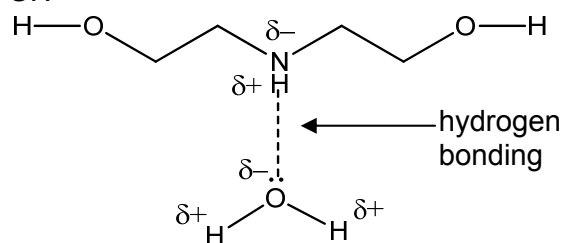
2nd (✓): Intramolecular hydrogen bonding observed in DEA will render intermolecular hydrogen bonding between DEA molecules to be less extensive. More energy is required to overcome the more extensive hydrogen bonds formed between TEA molecules than those between DEA molecules.

[1m]

2. (c) (ii)



OR



[1m]

2. (d) (i) $\text{pOH} = 14 - 9.6 = 4.4$

$$[\text{OH}^-] = 10^{-4.4} = 3.98 \times 10^{-5} \text{ mol dm}^{-3} \quad [1\text{m}]$$

$$[\text{Cd}^{2+}] \text{ in mol dm}^{-3} = \frac{K_{\text{sp}}}{[\text{OH}^-]^2} = \frac{2.5 \times 10^{-14}}{(3.98 \times 10^{-5})^2} = 1.58 \times 10^{-5} \text{ mol dm}^{-3}$$

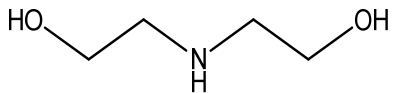
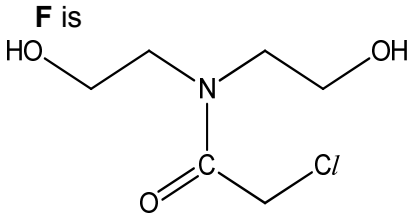
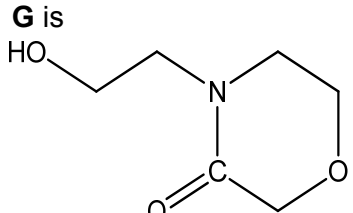
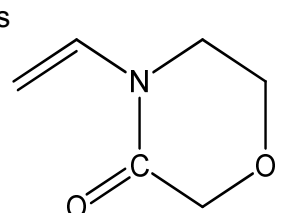
$$[\text{Cd}^{2+}] \text{ in g dm}^{-3} = 1.58 \times 10^{-5} \times 112 = \underline{1.77 \times 10^{-3} \text{ g dm}^{-3}} \quad [1\text{m}] \text{ 3sf}$$

2. (d) (ii) $[\text{Cd}^{2+}] = \frac{1.77 \times 10^{-3} \text{ g}}{1 \text{ dm}^3} = \frac{1.77 \times 10^{-3} \times 10^6 \mu\text{g}}{1 \times 10^3 \text{ cm}^3} = \underline{1.77 \mu\text{g cm}^{-3}}$

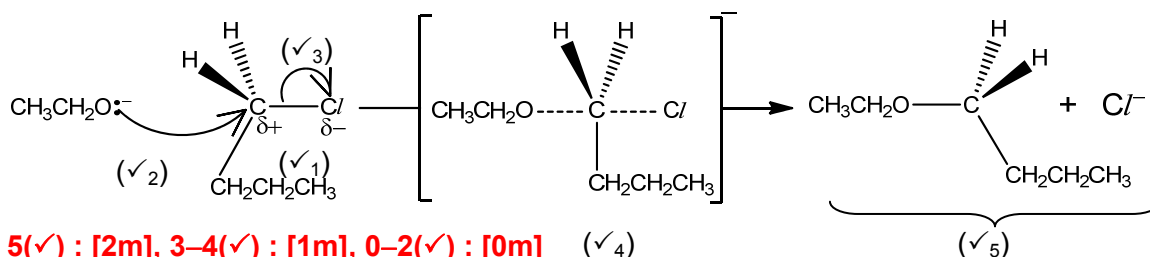
Vegetable produced will not be safe for human consumption.

[1m]

2. (e)

Reactions	Deductions	
 $+ \text{ClCH}_2\text{COCl} \rightarrow \text{F}$ 1 mol 1 mol neutral <u>Condensation / Nucleophilic substitution</u> (✓)	F is neutral. F cannot be an amine. F is an <u>amide</u>.	
F + Na $\xrightarrow{\text{warm}}$ cyclic G, $\text{C}_6\text{H}_{11}\text{NO}_3$ 1 mol 1 mol <u>Acid-metal reaction / redox</u> (✓) followed by <u>Nucleophilic substitution</u> (✓)	F is an <u>alcohol</u> undergoing acid-metal reaction to give alkoxide. The alkoxide formed acts as a nucleophile to attack alkyl halide in F via intramolecular nucleophilic substitution. G is a cyclic <u>ether</u>.	
G + $\text{Al}_2\text{O}_3 \rightarrow \text{H}$ <u>Elimination / dehydration</u> (✓)	G is an <u>alcohol</u>.	
H + $\text{Br}_2 \rightarrow$ decolourises <u>Electrophilic addition</u> (✓)	H is an <u>alkene</u>.	
4–5 (✓) : [2m]; 2–3 (✓) : [1m]		
F is  [1m]	G is  [1m]	H is  [1m]

3. (a)

3. (b) (i) **Na** [1m]

3. (b) (ii) The p-p orbital overlap results in the **delocalisation of the lone pair** of electrons **on O** atom **into the benzene ring** of the phenoxide, **making the phenoxide less able to act as a nucleophile**. Hence phenoxide is a weak nucleophile. [1m]

3. (b) (iii) The **rate of reaction involving iodoethane** will be **faster/higher**. [1m]

From the *Data Booklet*,

$$E(\text{C}-\text{Cl}) = 340 \text{ kJ mol}^{-1}; E(\text{C}-\text{I}) = 240 \text{ kJ mol}^{-1}$$

OR Since the bond energy of **C–I bond** < of **C–Cl bond**,

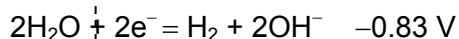
C–I bond is weaker than C–Cl bond and thus **requires less energy to break the C–I bond**. [1m]

OR

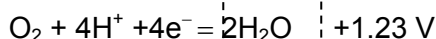
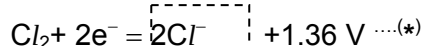
Atomic radius of Cl = 0.099 nm ; Atomic radius of I = 0.133 nmOR I atom is bigger (in size) than Cl atomC-I bond length is longer and weaker than C-Cl bond and thus requires less energy to break the C-I bond. [1m]

3. (c) (i)
- Concentrated KCl/contains: K^+ , Cl^- , H_2O

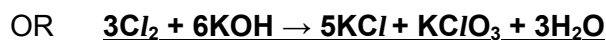
At the cathode:

Since $E'(H_2O/H_2)$ is more positive(or less negative) than $E'(K^+/K)$, H_2O will be reduced in preference over K^+ . Hence producing H_2 and OH^- at the cathode. [1m]

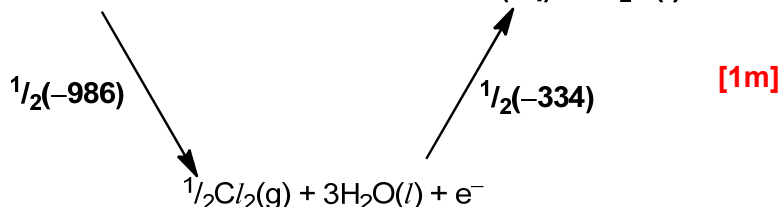
At the anode:

In concentrated KCl, $[Cl^-]$ is high/increases, which results in the equilibrium position of (*) to shift to the left. [1m]This makes the $E(Cl_2/Cl^-)$ is more negative/ less positive than $E'(O_2/H_2O)/+1.23 \text{ V}$, Cl^- will be oxidised in preference over H_2O , hence producing Cl_2 at the anode. [1m]

3. (c) (ii)
- Cl_2
- undergoes disproportionation with
- hot
- KOH/ OH^-
- to form
- ClO_3^-
- .
- [1m]



3. (c) (iii)
- $ClO_3^-(aq) + 6e^- + 6H^+(aq) \xrightarrow{\Delta H_f^\circ} Cl^-(aq) + 3H_2O(l)$
- [1m]

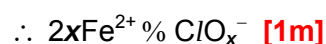
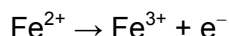


$$\Delta H^\circ = \frac{1}{2}(-986) + \frac{1}{2}(-334) = \underline{-660 \text{ kJ mol}^{-1}} \quad [1m]$$

3. (c) (iv)
- $-841\,000 = - (6) 96500 (E^\circ)$

$$E^\circ(ClO_3^-/Cl^-) = + \underline{1.45 \text{ V}} \quad [1m]$$

3. (d) (i)
- $ClO_x^- + 2xe^- \rightarrow Cl^-$

OR mole ratio of $ClO_x^- : Fe^{2+}$ is
1 : 2x

3. (d) (ii) Amount of Fe^{2+} used = $\frac{21.0}{1000} \times 0.540 = 0.0113 \text{ mol}$ [1m]

$$\text{Amount of } \text{ClO}_x^- = \frac{0.0113}{2x} \text{ mol}$$

$$\text{Amount of } \text{KClO}_x = \frac{0.302}{39.1 + 35.5 + x(16)} \text{ [1m]}$$

$$\text{Amount of } \text{ClO}_x^- = \frac{0.302}{39.1 + 35.5 + x(16)} = \frac{0.0113}{2x}$$

$$\frac{2(0.302)x}{0.0113} = 74.6 + x(16)$$

$$53.5x - 16x = 74.6$$

$$x = 1.99 \approx \underline{2} \text{ [1m]}$$

4 (a) (i) $K_c = \frac{[\text{SO}_2][\text{Cl}_2]}{[\text{SO}_2\text{Cl}_2]} \text{ [1m]} \quad \underline{\text{mol dm}^{-3}} \text{ [1m]}$

4 (a) (ii) Total amount of H^+ in 25.0 cm^3 of soln = $\frac{38.50}{1000} \times 5.00 = 0.1925 \text{ mol}$

$$\text{Total amount of } \text{H}^+ \text{ in } 250 \text{ cm}^3 \text{ of soln} = \frac{250}{25.0} \times 0.1925 = 1.925 \text{ mol}$$

$$\therefore \text{Amount of } \text{SO}_2\text{Cl}_2 \text{ in the original equilibrium} = \frac{1}{4} \times 1.925 = \underline{0.481 \text{ mol}} \text{ [1m]}$$

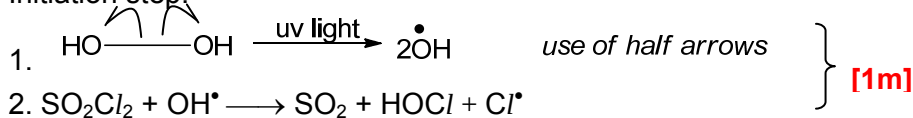
	$\text{SO}_2\text{Cl}_2(\text{g})$	=	$\text{SO}_2(\text{g})$	+	$\text{Cl}_2(\text{g})$
Initial amount/ mol	1.0		0		0
Change/ mol	-0.519		+ 0.519		+ 0.519
Eqm amt/ mol	0.481		<u>0.519</u>		<u>0.519</u> [1m]

$$K_c = \frac{\left(\frac{0.519}{3.60}\right)^2}{\left(\frac{0.481}{3.60}\right)} \text{ [1m]}$$

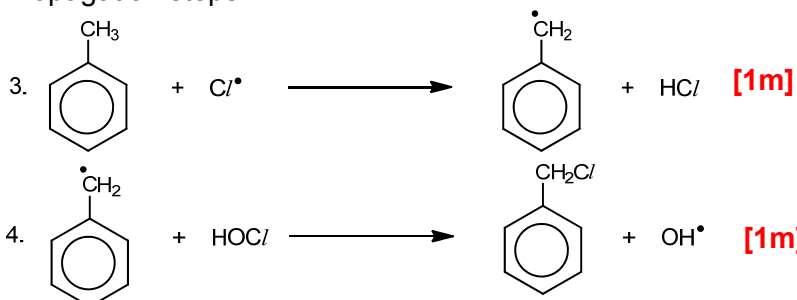
$$= \underline{0.156 \text{ mol dm}^{-3}} \text{ [1m]}$$

4 (a) (iii) When temperature is increased, the equilibrium position would shift to the right to favor the endothermic reaction, to absorb some heat so as to lower the temperature. [1m]

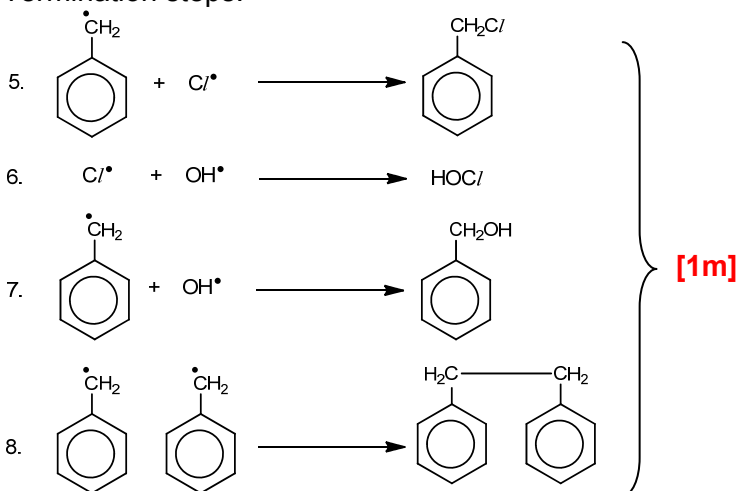
(b) Initiation step:



Propagation steps:

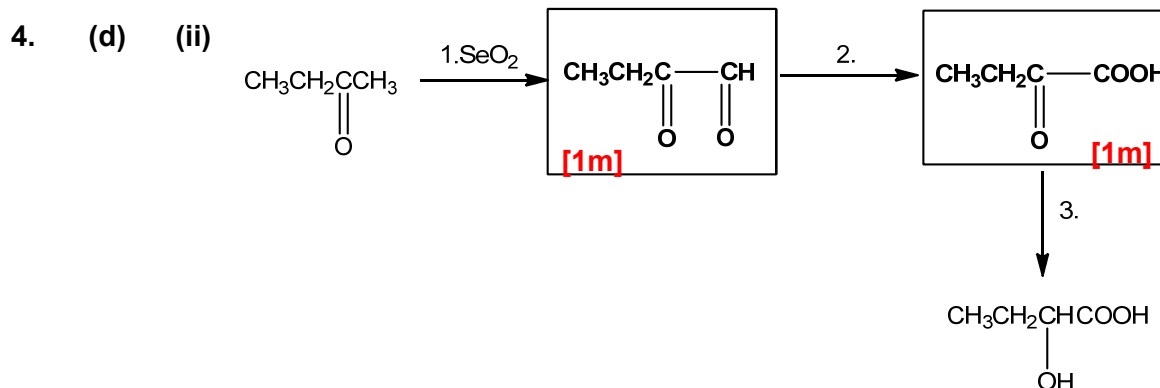


Termination steps:

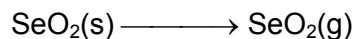


4. (c)
$$\underline{A_r} \text{ of Se} = \frac{\{(0.9 \times 74) + (9.0 \times 76) + (7.6 \times 77) + (23.5 \times 78) + (49.8 \times 80) + (9.2 \times 82)\}}{100}$$

$= \underline{79.1}$ [1m]

4. (d) (i) **Oxidising** agent [1m]2. **acidified KMnO₄ / K₂Cr₂O₇, heat under reflux** [1m]3. **H₂, Ni** catalyst, **heat** (or **NaBH₄**, methanol) [1m]

4. (e)



$\Delta H > 0$ as energy is required to overcome the intermolecular forces between SeO_2 molecules [1m]

$\Delta S > 0$ as there is an increase in disorder as the reaction proceeds with an increase in number of gaseous molecules, OR

from a more ordered solid state to form a less ordered gaseous state. [1m]

When temperature increases, $-\text{T}\Delta S_r'$ becomes increasingly negative.

$$\Delta G = \underbrace{\Delta H_r'}_{+ve} - \underbrace{\text{T}\Delta S_r'}_{-ve}$$

① When T increase, $-\text{T}\Delta S_r'$ becomes more negative.

At high T, $|\text{T}\Delta S_r'| > |\Delta H_r'|$ (✓)

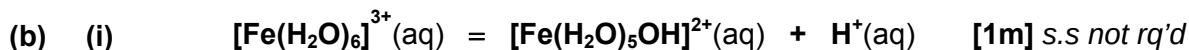
② and thus $\Delta G < 0$ (or ΔG becomes negative) (✓).

Hence sublimation of SeO_2 is spontaneous at higher temperatures.

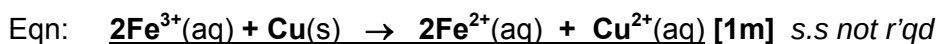
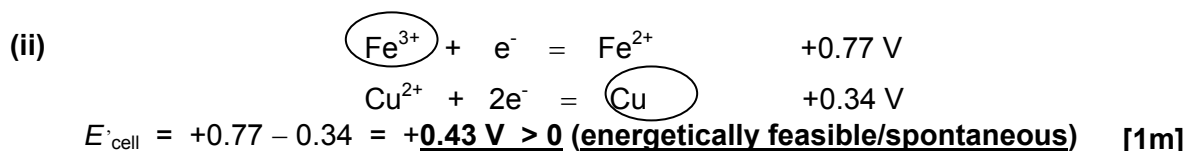
2(✓) : [1m]

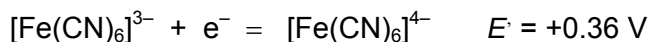
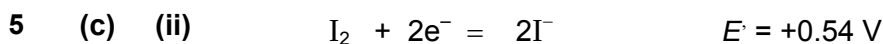
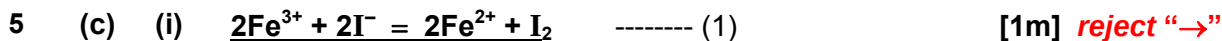
5. (a) Iron can exist in oxidation states +2, +3 and +6 because variable number of electrons from 3d and 4s orbitals can be involved in bonding since 3d and 4s orbitals have similar energies. [1m]

Calcium cannot form oxidation state higher than +2 because this involves the removal of electron from an inner quantum shell (3p) which would require too much ionisation energy. [1m]



- 1st ✓: Due to the high charge density of Fe^{3+} ,
 2nd ✓: the hydrated complex of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ can undergo hydrolysis as Fe^{3+} has high polarising power to polarise H_2O ligand, which weakened the O-H bond of the water ligand, hence producing H^+ ions [1m]

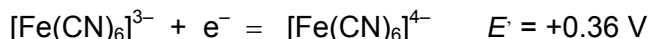
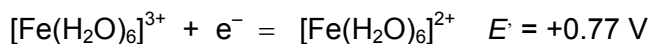




Since $E^\circ(\text{Fe}(\text{CN})_6^{3-}/\text{Fe}(\text{CN})_6^{4-})$ is less positive than $E^\circ(\text{I}_2/\text{I}^-)$, it implies that $\text{Fe}(\text{CN})_6^{4-}$ is preferred to be oxidised and I_2 be reduced. [1m]

Hence, position of equilibrium in (c)(i) will shift to the left. [1m]

Alternatively:



$E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+})$ is more positive than $E^\circ(\text{Fe}(\text{CN})_6^{3-}/\text{Fe}(\text{CN})_6^{4-})$, the oxidising power of Fe(III) decreases when H_2O ligands are replaced by CN^- ligands. [1m]

Hence, the equilibrium position in c(i) will shift to the left. [1m]

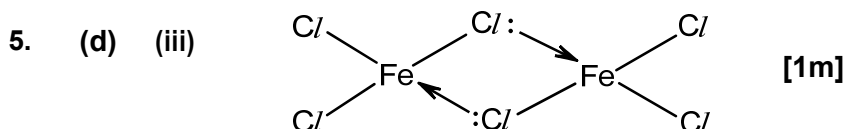
5. (d) (i) $M_r = \frac{mRT}{PV}$

Apparent $M_r = \frac{1.00 \times 8.31 \times (500 + 273)}{26.3 \times 10^3 \times 1.00 \times 10^{-3}} = \underline{244}$ or 244.2 [1m]

5. (d) (ii) Let the percentage of the dimer be x %.

Apparent $M_r = \frac{(100 - x) \times 162.3 + (x \times 324.6)}{100} = 244$

$x = \underline{50.3 \%}$ [1m] 3 s.f.



5 (e) The quaternary structure of haemoglobin refers to the spatial arrangement of 4 polypeptide chains (or 2 α chains and 2 β chains) [1m]

stabilised by R-group interactions : ionic interactions, hydrogen bonds and van der Waals' interactions. [1m]

5. (f) (i) Dative covalent bond/ co-ordinate bond [1m]

5. (f) (ii) As body cells use up oxygen for essential metabolic processes, $[\text{O}_2]$ decreases and this will shift the equilibrium position of $\text{Hb}(\text{aq}) + 4\text{O}_2(\text{g}) = \text{Hb}(\text{O}_2)_4(\text{aq})$ to the left, causing oxyhaemoglobin to release free oxygen. [1m]

5. (g) Oxidation state of Cu in oxygenated state / blue haemocyanin = +2
 Oxidation state of Cu in deoxygenated state / colourless haemocyanin = +1 } [1m]

In the oxygenated state, haemocyanin is coloured because Cu²⁺ (d⁹) has a partially filled 3d orbital. Thus electron transition between 3d orbitals is possible. [1m]

In the deoxygenated state, haemocyanin is colourless because

- 1st ✓: electron transition between 3d orbitals is not possible
 2nd ✓: since all 3d orbitals are completely filled / Cu⁺ has d¹⁰ configuration / there is no partially filled/unfilled/vacant 3d orbitals } [1m]

